OOP
Functional Programming

JS	sort(callback)	map,filter,
Java	sort(Comparator c)	Stream

TS

```
interface Comparable<T>{
        int compareTo(T obj)
}
```

```
class Employee implements Comparable{

        int compareTo(Object obj){

}
}
```

```
Employee e1 = new Employee()
Employee e2 = new Employee();
e1.compareTo("sunbeam")
```

Java		Typescript	
- Compiled	javac	- compiled	tsc
- Interepreted		- interpreted	
.class		.js	

Runtime Environment	Runtime Environment
JRE	Browser Node

javac Program01.java	tsc demo01.ts
javac -d ..\bin Program01.java	tsc --outDir ..\output demo01.ts
change directory to bin	change the directory to output
java Program	node demo01.js

```
Object[] arr = new Object[5];
arr[0] = 10;
arr[1] = "subeam"
```

Types in TypeScript

Special Types
- void
- never
- unknown
- any

Object Types
- Array
- Tuple
- Object

#Advanced Types
- type (type alias)
- union
- intersection
- interface

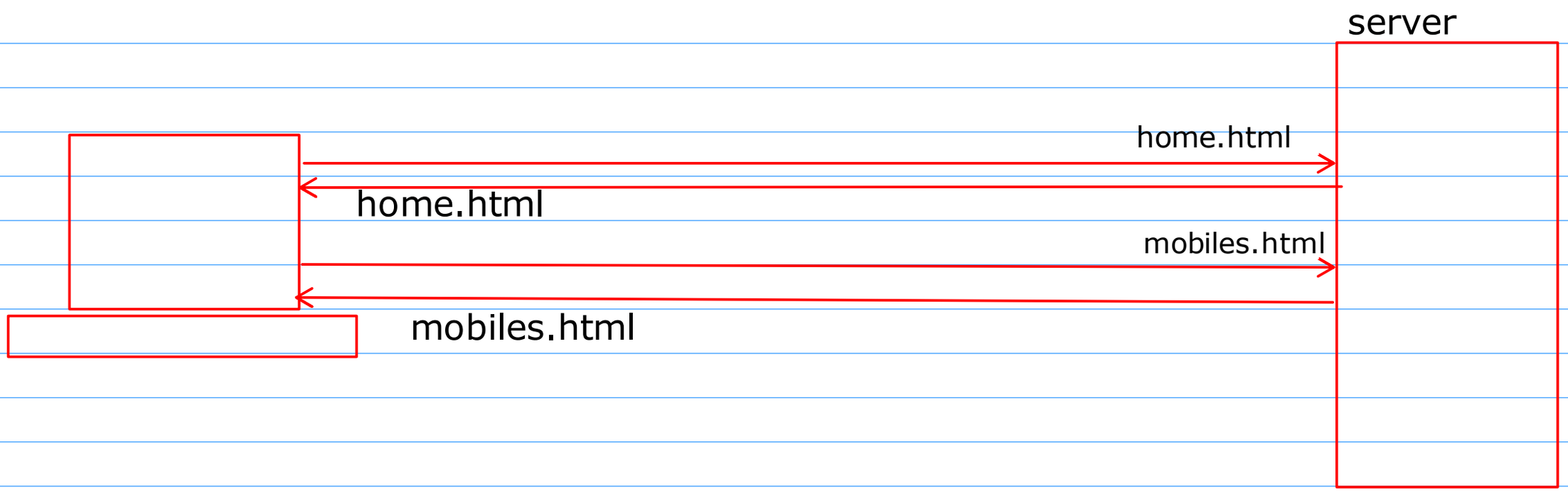
HTML
CSS
JS
Node
Express
TypeScript
React

Console Based
CPP
C
Java
Python
JS
TS

Web Application
HTML
CSS
JS
React

Backend Application
JS -> Express
Java -> Spring

React , Angular
Single Page Applications
Multi Page Applications



server

client

index.html

index.html

document

document

html

head

body

meta

title

h1

div root

h1

div

p

p

h2

p

html

head

body

meta

title

h1

div root

p

p

React

virtual DOM

Diffing

SPA

h1

h2

p

p

h1

div

p

p

Memory Areas Virtual Memories

Method Area

Java Stack

Java Heap

.....

div-OG

li li li li

div-V

li li li li

JVM->Virtual Machine

og_div.innerHTML = div_v.innerHTML

h1

div

p

p

JSX -> Java Script and XML